However, the number of free electrons provided by the pentavalent impurity far exceeds the number of holes. It is due to this predominance of electrons over holes, hence it is called n-type semiconductors (n-stands for negative).

3.9 P-TYPE SEMICONDUCTORS

When a small amount of trivalent impurity is added to a pure semiconductor, it is called p-type semiconductor.

The addition of trivalent impurity provides a large number of holes in the semiconductor. Typical examples of trivalent impurities are Gallium (Atomic No: 31) and Indium (Atomic No: 49). Such impurities produce p-type semiconductors are known as acceptor impurities, because the holes created can accept the electron.

In Figure 3.24, Gallium is added with Germanium crystal to form p-type semiconductor. Each atom of gallium fits into the germanium crystal. But, only three covalent bonds can be formed. The fourth bond is incomplete, being short of one electron. The missing electron is called a hole. Therefore, for each Gallium atom added, one hole is created. A small amount of Gallium provides millions of holes.

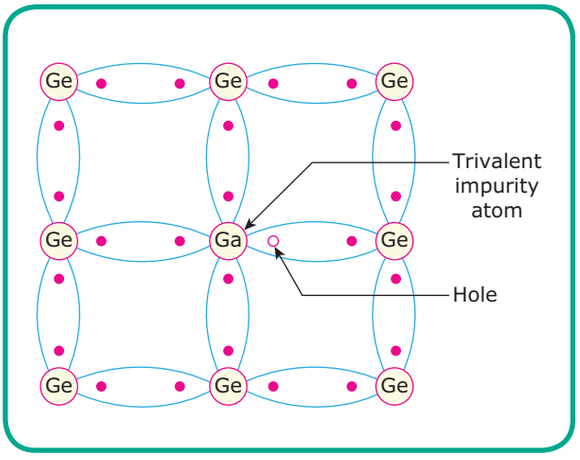


Figure 3.24 P-type semiconductor

Hence, in p-type semiconductor, holes are the majority carriers. When potential difference is applied to the p-type semiconductor the holes are shifted from one covalent bond to another. As the holes are positively charged, they are directed towards the negative terminal, constituting what is known as "hole current". The p-type semiconductor and its band structure are shown in Figure 3.25.

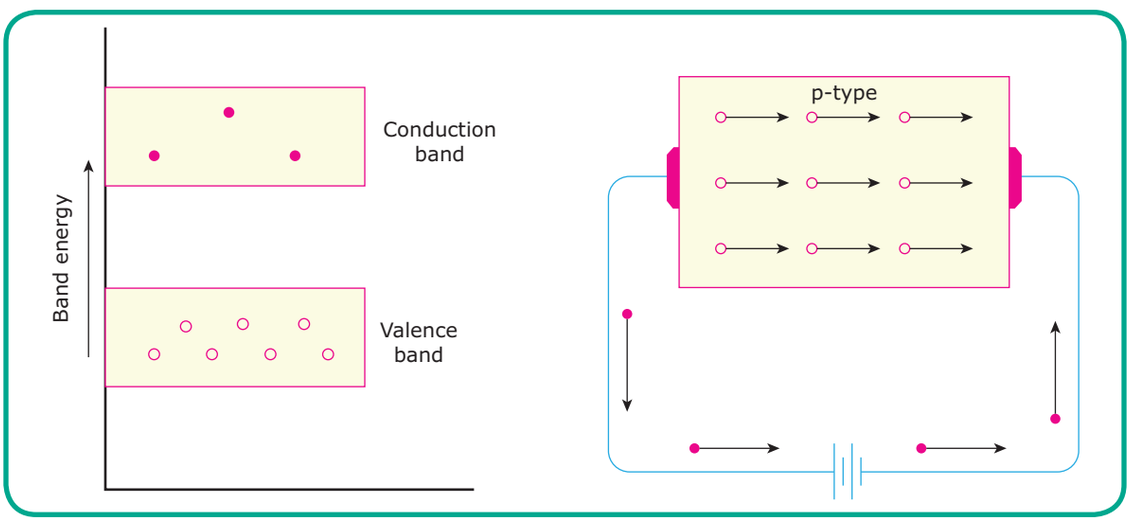


Figure 3.25 Energy band and current flow of P-type semiconductor

ELEC_EQUIP_Chapter03.indd 77 11/10/2020 11:36:42 AM